MILITARY SPECIFICATION SHEET

INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-14A

VOX MODULE

Inactive for new design after 10 August 1984
For new design use MIL-I-22352/7(AS).

This specification sheet is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the unit described herein shall consist of this document and the latest issue of specification MIL-I-22352(AS).

1. SCOPE

1.1 Scope.

This specification sheet provides performance and physical characteristics for the VOX Module used in active Interphone Control panel C-2645C/AIC-14.

1.2 Military Part Number.

The military part number shall be M22352/6-1

2. APPLICABLE DOCUMENTS

Refer to 20 of MIL-I-22352(AS) Appendix A for a listing of applicable documents. The requirements specified herein shall have precedence over all reference documents. The contract and its attachments shall have precedence over any specification or reference document.

3. REQUIREMENTS

3.1 Item Definition. The VOX Module shall provide voice activated transmitter key relay activation characteristics as specified herein when used in master Interphone Control panel C-2645C/AIC-14.

FSC 5831

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3.1.1 Item Diagram. The VOX Module shall provide the functions represented schematically in accordance with the diagram of Figure 1.

3.1.2 Interface Definition. The VOX Module input and output signals will normally display the characteristics presented in Table I.

TABLE I. Signal characteristics.

Signal Name	Characteristics
Inputs:	
VOX (VOX)	Voice range audio (300 to 6000 Hz) signal ranging in amplitude from 0 to 18 Vac. No dc component normally present.
VOX SENS Center (VOX SENS CTR)	Voice range audio (300 to 6000 Hz) signal ranging in amplitude from 0 to 6 Vac. No dc component normally present.
VOX SENS Low (VOX SENS LO)	Voice range audio (300 to 6000 Hz) signal ranging in amplitude from 0 to 0.1 Vac. No dc component normally present.
Power (PWR)	+27.5 ±0.5 Vdc
Outputs:	
VOX SENS High (VOX SENS HI)	Voice range audio (300 to 6000 Hz) signal ranging in amplitude from 0 to 18 Vac. No dc component normally present.
VOX (VOX)	Voice range audio (300 to 6000 Hz) signal ranging in amplitude from 0 to 1 Vac. No dc component normally present.
Grounds:	
Power (PWR GND)	Power and Signal return. See 3.4.4.2 of MIL-I-22352(AS).

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3.2 Characteristics

3.2.1 Performance Characteristics

3.2.1.1 Input Characteristics

3.2.1.1.1 VOX (VOX). The audio signal from the microphone (or microphone circuitry) shall be applied to the VOX input. The VOX input shall be connected to the VOX SENS HI output via an internal resistor. The VOX input shall also be applied to an internal relay which shall feed the VOX output when the VOX audio exceeds the threshold limit as defined herein.

3.2.1.1.2 VOX Center (VOX CTR). The VOX Module shall have an VOX CTR input for activation of a relay based on the input audio amplitude. The input to the relay closure sensing circuit shall have the following characteristics:

3.2.1.1.2.1 Frequency. The VOX relay sensing circuitry shall operate in response to inputs over the frequency range from 300 to 6000 Hz.

3.2.1.1.2.2 Amplitude. The VOX CTR input shall accept signals within the specified frequency range which have an amplitude which range from 0 to 1.0 volt. When installed in the C-2645C/AIC-14, the amplitude of the input will be determined by the level of the audio source, the gain of the microphone circuitry, and by the VOX SENS control internal to the C-2645C/AIC-14 but external to the VOX Module.

3.2.1.1.3 VOX SENS Low (VOX SENS LO). The VOX SENS LO signal is the return side of the microphone audio input signal from the VOX SENS control. The VOX SENS LO signal shall be routed to ground via a resistor of appropriate value.

3.2.1.1.4 Power (PWR). The VOX Module shall accept and operate within the constraints of this specification when supplied with 27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1.

3.2.1.1.5 Power (PWR). The VOX Module shall accept and operate within the constraints of this specification when supplied with 27.5 ±0.5 Vdc electrical power as shown in the diagram of Figure 1.

3.2.1.2 Output Characteristics. The VOX Module shall have the following output characteristics:

3.2.1.2.1 VOX SENS High (VOX SENS HI). The VOX SENS HI output signal shall be derived from the drive side of the microphone input audio (VOX) signal. When installed, the VOX SENS HI output is applied to the VOX SENS control which is external to the VOX Module.

3.2.1.2.2 VOX (VOX). The VOX Module shall have a VOX output which shall have two output states (1) Open and (2) an audio signal taken from the VOX input (less than 2 dB insertion loss). The VOX output shall present an open (greater than 40 dB isolation) circuit for VOX signals from 300 to

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6000 Hz having input amplitudes which are below the specified threshold value. Once the specified threshold value has been met the VOX output shall change state to an audio output. Any subsequent decrease in the VOX input amplitude of more than 10 percent below the threshold value shall produce a change in state of the VOX output to an open circuit. The VOX module, when operated in the circuit of Figure 3, shall produce an audio output for an input at TP1 which is not more than 1.4 Vac with resistor R1 set to maximum resistance. When resistor R1 is set to minimum resistance, the VOX output shall produce an audio output when the input amplitude at TP1 is increased to 12.0 Vac. The attack time shall not exceed 15 milliseconds after the application of the first positive half cycle when the input has an amplitude of 5 dB above the threshold value. The release time shall be between 100 and 500 milliseconds. VOX switching shall be stable. The VOX output shall not oscillate or chatter when the input is maintained at any constant level near threshold.

The following definitions shall be used in applying the VOX response criteria.

3.2.1.2.1.1 Threshold. The VOX threshold is defined as the level required to energize the key relay, producing an audio output taken from the VOX input as shown in Figure 1.

3.2.1.2.1.2 Attack Time. Attack time is defined as the time required to close the relay following application of a 1,000 Hz signal to the VOX input which meets or exceeds the sensing circuit threshold.

3.2.1.2.1.3 Release Time. Release time is defined as the time required to open the output circuit after the removal of the 1,000 Hz signal at the VOX input.

3.2.1.2.1.4 Near Threshold. For the purposes of this specification near threshold is defined as within ±2 percent of threshold value.

3.2.2 Physical Characteristics

3.2.2.1 Dimensions. The VOX Module shall consist of a single encapsulated module having the dimensions of Figure 2. All outside dimensions shall be within ±0.03 inches of the nominal value shown.

3.2.2.2 Weight. The amplifier shall weigh no more than 2.0 ounces.

3.3 Design and Construction. The design and construction of the amplifier shall be in accordance with the requirements of 3.3 of MIL-I-22352(AS).

4. QUALITY ASSURANCE PROVISIONS

See MIL-I-22352(AS).

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4.1 Responsibility for Tests and Inspections. See MIL-I-22352(AS).

4.2 Test plan. A detailed test plan will be submitted by the contractor for approval by the acquiring activity which details the tests to be performed in order to verify compliance with the performance characteristics specified in 3.2.1. The test plan will consider all of the inspections delineated in 4.2 of MIL-I-22352(AS).

4.3 Test procedures. Detailed test procedures will be submitted by the contractor for approval by the acquiring activity which will satisfy the requirements of the test plan of 4.2.

Preparing Activity:
Navy - AS

Project No. 5831-N042-6

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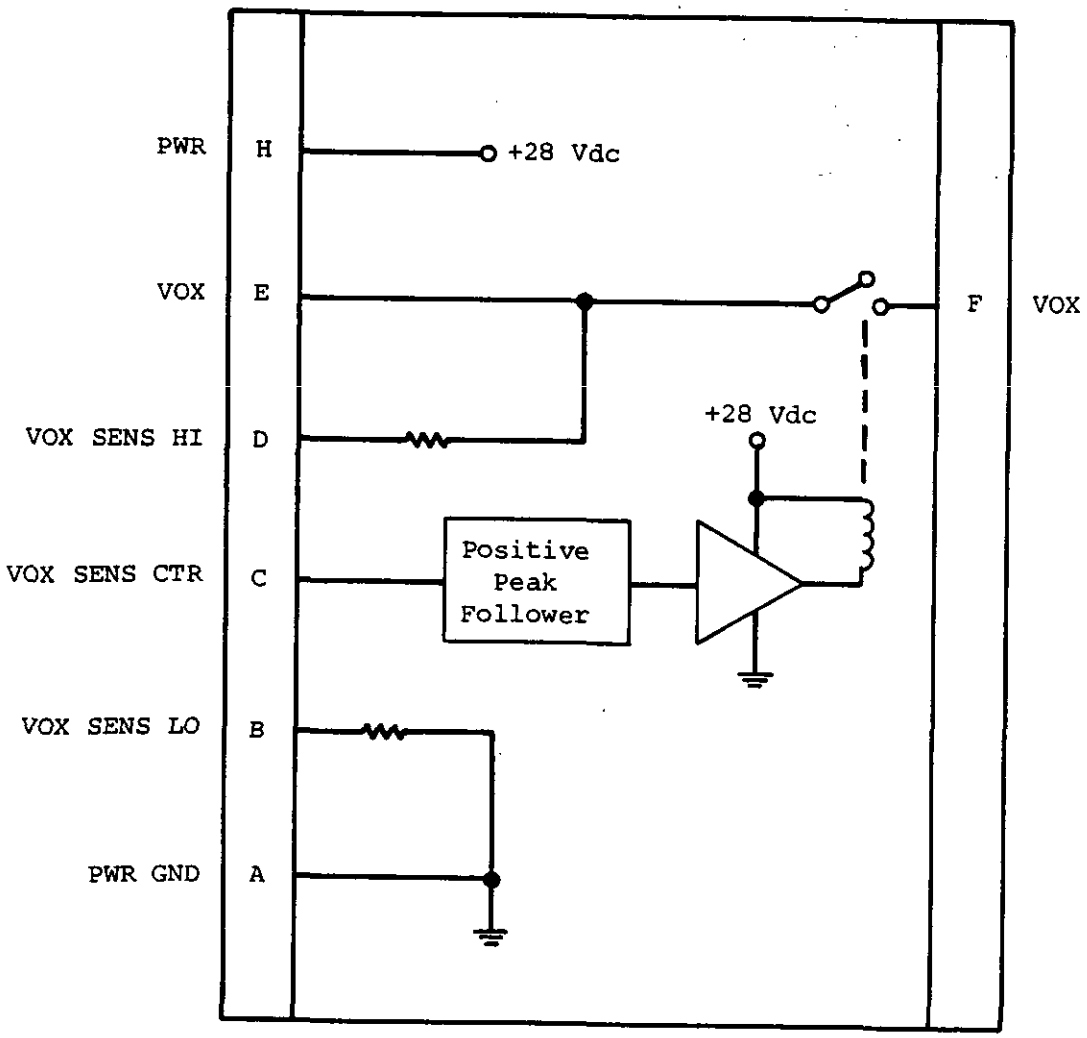
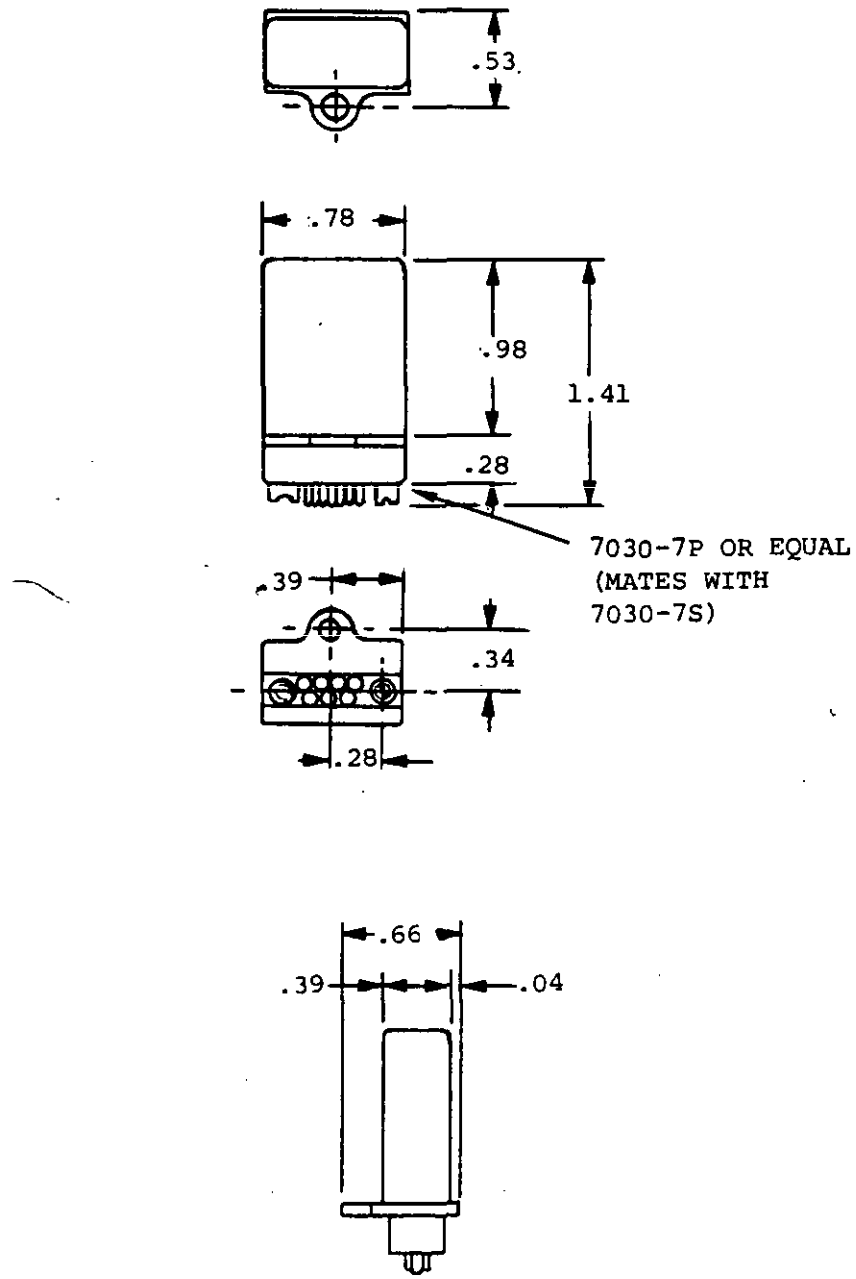


FIGURE 1. VOX module functional diagram.

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1/ All dimensions are in inches. Tolerances on dimensions are ±0.03 inches unless otherwise specified.

FIGURE 2. VOX module outline drawing.

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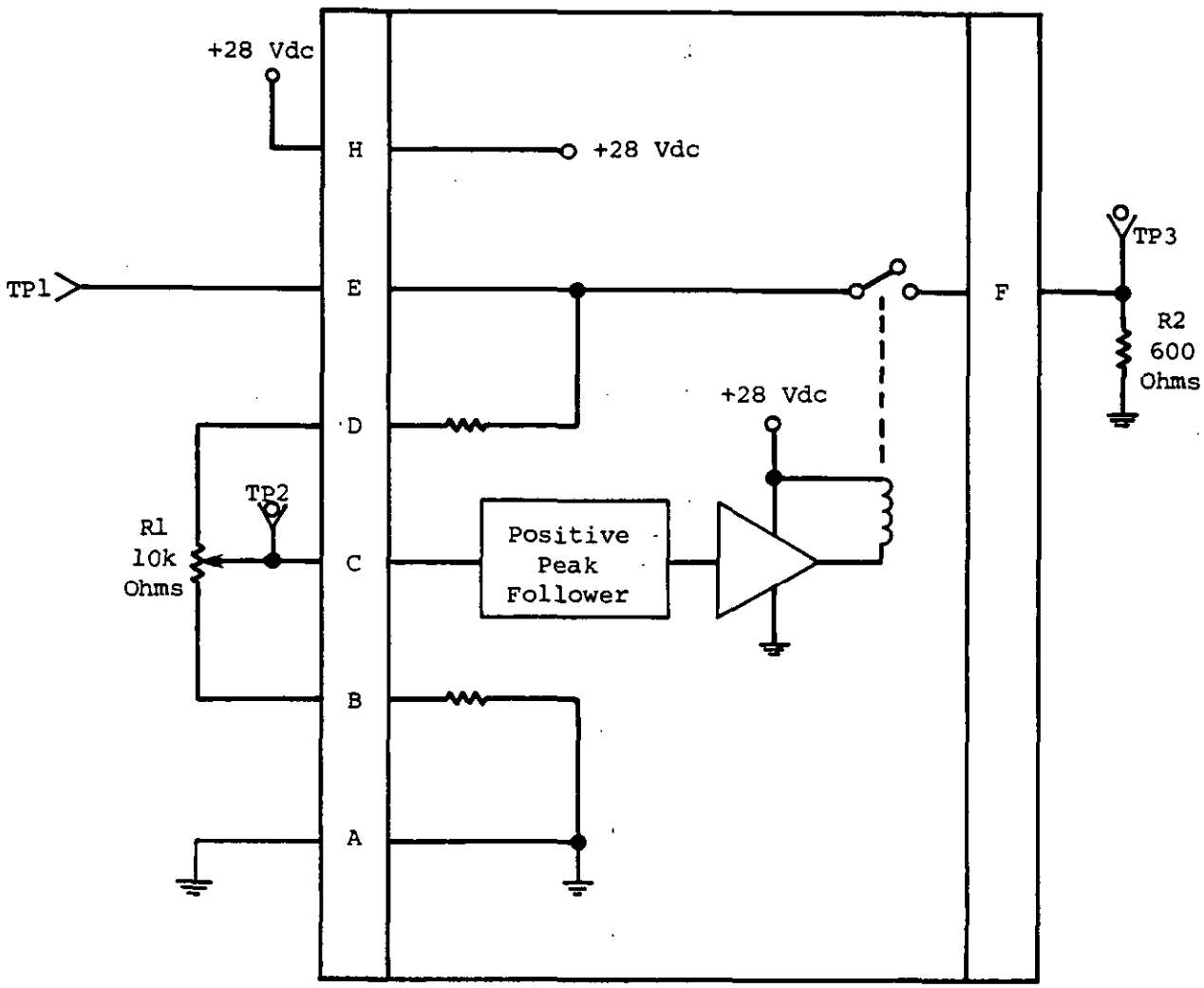


FIGURE 3. VOX module test set-up.

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions - Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-22352/6(AS)	INTERCOMMUNICATION SETS, AN/AIC-14 AND AN/AIC-14A VOX MODULE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
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5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
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